

Fucheng Liu (Eric)

E1303008@u.nus.edu | Singapore
linkedin.com/in/fucheng-liu-eric

EDUCATION

National University of Singapore | Aug 2023 - May 2027 (expected)

B.Sc., Mathematics; Second Major in Quantitative Finance; Minor in Data Science and Analytics.

Selected coursework represented in projects: Discrete Optimisation (MA4254), Regression Analysis (ST3131), Artificial Intelligence: Technology and Impact (IT1244).

Shanghai Jiao Tong University | Summer Session, Jun - Jul 2025

Algebra and Number Theory (A); Complex Variable (A-).

RESEARCH INTERESTS

Financial time series, multimodal learning, model uncertainty, and optimization.

MANUSCRIPTS UNDER REVIEW

Fucheng Liu, Chenxue Yang. *Adaptive Temporal-Context Fusion for Policy-Aware Multimodal Agricultural Stock Movement Prediction*. Under review at *Expert Systems with Applications*, 2026. Not yet accepted for publication.

RESEARCH EXPERIENCE

Policy-Aware Multimodal Stock Movement Prediction

- Investigated the fusion of historical price features, macroeconomic policy information, and corporate announcements for agricultural stock movement prediction.
- Contributed experimental design, implementation, debugging, comparative experiments, manuscript revision, and initial drafting.
- Used Python for data processing, model training, and evaluation; examined the role of temporal context in multimodal prediction.

SELECTED INTERNSHIP EXPERIENCE

China Merchants Securities | Quantitative Research Intern | May - Aug 2026

- Replicated and tested academic factors, including Tail Risk Exposure and Skewness Dispersion, using daily A-share market data, portfolio backtesting, and risk-return evaluation.
- Studied corporate events such as private placements using average abnormal returns, hit rates, and price changes around announcements.
- Used Python (Pandas, NumPy, Matplotlib), Wind, and iFinD for data processing and analysis, and prepared research reports and statistical exhibits.

Avalok Capital | Quantitative Research Intern (Remote) | Oct - Dec 2025

- Worked on replication and backtesting of formulaic alpha signals and evaluation of multi-factor models.
- Performed exploratory data analysis and factor correlation studies to examine redundancy and behavior across market conditions.

SELECTED PROJECTS

Optimization-Based Group Testing | MA4254 Discrete Optimisation

- Formulated sparse recovery in non-adaptive group testing as an integer linear program with binary variables and test-consistency constraints.
- Implemented ILP and LP-relaxation algorithms with randomized rounding, alongside rule-based DNI/DI estimators.
- Compared exact recovery probability and runtime in simulations with up to 500 individuals under varying testing budgets.

HDB Resale Price Prediction | ST3131 Regression Analysis

- Analyzed Singapore HDB resale transactions using location, flat type, floor area, storey range, and remaining lease as predictors.
- Compared multiple linear regression specifications using variable selection, significance testing, and model diagnostics; interpreted quantitative and categorical effects.

Biological Sequence Classification | IT1244 Artificial Intelligence

- Collaborated in a three-person team on labelled DNA and protein sequence classification.
- Contributed to feature extraction, preprocessing, model training and comparison, and accuracy evaluation on an unseen test set.

ADDITIONAL EXPERIENCE

October Capital | Venture Capital Investment Intern | May - Jul 2025

- Conducted market research and financial analysis supporting due diligence for three companies and investment committee discussions.

Volkswagen (Anhui) Automotive | Productivity Management Intern | May - Jul 2024

- Supported operational data tracking, workflow coordination, and cross-department process improvement.

SELECTED COMPETITIONS

Moomoo Options Paper Trading Competition | Jan 2026

Ranked 2nd among 5,432 participants in simulated options trading.

Optiver Trading Simulation Challenge | Apr 2025

Ranked in the top 10% among approximately 100 participants; implemented a grid trading strategy in a simulated environment.

TECHNICAL SKILLS

Programming and analysis: Python (NumPy, Pandas, Matplotlib), R, MATLAB, C++, Excel.

Financial data: Wind, Tonghuashun iFinD; time-series analysis, factor research, backtesting.

Mathematical background: Probability, stochastic processes, linear algebra, linear and nonlinear optimization, numerical methods.

Languages: English (fluent), Mandarin Chinese (native), Japanese (beginner).

SERVICE

St. Luke's Elder Care volunteer, Jan - May 2026; science communication activities with Singapore Science Centre through NUS SP1541, Jan - May 2025.